

Практикум D3

62 marks of the archive, six techniques, eleven tasks. Everything the archive asks about $X \sim B(n, p)$ from May 2021 to May 2025. All of it is on Paper 2, and all of it is a calculator question — which is exactly why people lose marks here: the calculator does the sum and does not check what it was asked to sum.

О практикуме

Собран долями двух тем, потому что темы «биномиальное распределение» в разметке нет: 36 баллов его вопросов лежат в `statistics.probability` (доля 0.152 — это 36 из 237) и 35 в `statistics.discrete_random_variables` (0.393 — 35 из 89). Граница между двумя темами случайна: соседние части одного вопроса и даже одна часть в двух зональных копиях ноября 2023 размечены в разные.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d3-binomial

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D3 — The binomial distribution

62 marks of the archive, six techniques, eleven tasks. Everything the archive asks about $X \sim B(n, p)$ from May 2021 to May 2025. All of it is on Paper 2, and all of it is a calculator question — which is exactly why people lose marks here: the calculator does the sum and does not check what it was asked to sum.

The one idea

A binomial variable counts successes.

The same trial, repeated a fixed number of times, independently, with the same chance of success each time. X is how many successes.

Four conditions, and each one is a question you can ask of the story: is there a fixed n ? only two outcomes per trial? does p stay the same? does one

trial tell you nothing about the next? If all four are yes, then

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k},$$

$$k = 0, 1, \dots, n$$

— $p^k(1 - p)^{n-k}$ is one particular order of k successes and $n - k$ failures, and $\binom{n}{k}$ counts the orders.

And the idea that carries the marks

Your calculator has two buttons for this: **binompdf** gives $P(X = k)$, and **binomcdf** gives $P(X \leq k)$ — *at most*, and nothing else. Every other phrase in a question has to be turned into one of those two by you:

more than 6 is $X \geq 7$, which is $1 - P(X \leq 6)$. *Fewer than 49* is $X \leq 48$. *At*

least 10 is $1 - P(X \leq 9)$.

The calculator will happily compute $1 - P(X \leq 10)$ for *at least 10*. It is a correct number. It is the answer to a different question.

How the checks work

They do not know the answers. Each check is handed **the model and the event**, written the way the paper writes them:

```
flights = Bin(64, 0.0711930)
verify_binomial('4e', 0.169, P(flights > 6))
```

is "*your number: is it the probability that more than six of 64 flights are late?*" The check knows only $P(X = k)$, and adds it up over $k = 7, 8, \dots, 64$. np , $np(1 - p)$ and $1 - (1 - p)^n$ appear nowhere in it — the mean is a sum of $k P(X = k)$, and the least n is found by trying $n = 1, 2, 3, \dots$

In your own cells, the calculator is there too: `binompdf(n, p, k)` and `binomcdf(n, p,`

k), or `binomcdf(n, p, a, b)` for $P(a \leq X \leq b)$ the way a Casio does it.

When you are wrong the check says **how** — and above all, **which boundary**:

what you wrote	what the check says
$1 - P(X \leq 5)$ for <i>more than 6</i>	« $X > 6$ » starts at 7, and this counts 6 as well
$1 - P(X \leq 10)$ for <i>at least 10</i>	« $X \geq 10$ » includes 10 itself
$P(X \leq 10)$ for <i>exactly 10</i>	cdf where pdf was wanted
$P(X \leq 49)$ for <i>fewer than 49</i>	the boundary is off (in the condition)
the model with $1 - p$	success and failure are swapped
<i>at least one of the two</i> for <i>exactly one</i>	the case where both happen does not belong here
0.042	two significant figures: (M1)A0 without working

A number computed from a probability you rounded to three figures on the way is **accepted, with a remark** — the markscheme does the same.

Order of work

level	what it means	tasks
	the model is given; one value, then a run of values	1-3
	describe the distribution, or cut it with a condition	4-7
	a binomial on top of something else, or n unknown	8-11

Every task is a real past-paper question, cited.

100% of these marks are on a calculator paper, and the number is honest. $P(X \leq 6)$ for $B(50, 0.08)$ is seven terms nobody adds by hand, and the markscheme writes *use of tables* as a method. What the calculator cannot do is the translation — and that is where the method marks sit.

```

import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit.py
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *            # checks +
Bin, P(), Expect, Var, binompdf, binomcdf

language('en')               # this
notebook is in English, and so are the checks

p = symbols('p')              # the
probability of success, when it is the
unknown

# Bin(n, p) is  $X \sim B(n, p)$ . Comparisons on it
are events, and P() adds them up:
#      X = Bin(8, 0.7)
#      P(X == 6), P(X <= 6), P(X == 3, given=X
<= 4)
# Expect(X) and Var(X) are E(X) and Var(X) –
the letter E is taken by e = 2.718...

print('ready; sympy', sp.__version__)
X = Bin(8, 0.7)
print('the model:           ', X)
print('P(X = 6):           ', binompdf(8,
0.7, 6))
print('P(X <= 6):          ', binomcdf(8,
0.7, 6))
print('the same, as an event:',
float(sympify(P(X <= 6))))

```

Map of the six techniques

#	technique	you recognise it by	it reduces to
1	one value	<i>exactly k</i>	binompdf
2	a run of values	<i>at most, at least, more than, fewer than</i>	binomcdf, with the boundary moved by you
3	mean and variance	<i>expected number, Var, a variance given</i>	np , $np(1 - p)$, $a^2\text{Var}(X)$
4	a condition inside one distribution	<i>given that</i> — and both events are about the same X	$\frac{P(\text{overlap})}{P(\text{condition})}$
5	binomial on top of something	a <i>box</i> , a <i>competitor</i> , a <i>day</i> that succeeds when something inside it does	model inside, model outside
6	unknown n	<i>least number of trials, find n</i>	logs, or a table

Techniques 1 and 2 are one skill: reading an event off the words. Every later technique uses technique 2 somewhere,

which is why it has the most tasks. Technique 3 is the only one where the answer is not a probability. Techniques 4 and 5 are D2's conditional probability and tree, with the binomial supplying the numbers. Technique 6 turns the question around.

Part 1. Reading the event off the words

Theory: is it binomial, and one value

A basketball player makes a free throw with probability 0.7, and takes 8 of them. Fixed number of trials, two outcomes, the same p , independent throws — so the number made is

$$\begin{aligned} X \sim B(8, 0.7), \quad P(X = 6) &= \binom{8}{6} (0.7)^6 (0.3)^2 \\ &= 28 \times 0.117649 \times 0.09 \\ &= 0.296 \end{aligned}$$

On the calculator that is `binompdf(8, 0.7, 6)`.

Write the model down before the number. " $X \sim B(8, 0.7)$ " is the method mark. With it, a correct number scores both marks; without it, the markscheme says *"if no working shown, award (M1)A0"* for a two-significant-figure answer, and sometimes for a three-figure one.

Not everything that counts is binomial. Drawing cards without replacement changes p each time. "*Until the first success*" has no fixed n — that is D2's first success. The four questions above are worth asking every time, and the exam sometimes asks you to state them.

Where p comes from. In half the questions in this topic, p is the answer to an earlier part — a normal probability, an integral, a percentage. Carry it **unrounded**. $p = 0.365$ instead of 0.365112 changes the fourth figure of the answer, and the fourth figure is sometimes the third.

Task 1 ● — **May 2022 TZ1 Paper 2 Q11(b) and May 2023 TZ2 Paper 2 Q3(d), 4 marks**

(a) The weights of chocolate muffins are normally distributed, and part (a) of the question finds that the probability that a randomly selected muffin weighs less than 61 g is 0.365112...

In a random selection of 12 chocolate muffins, find the probability that exactly 5 weigh less than 61 g.

(b) The weights of bags of rice are normally distributed, and the earlier parts of the question find that the probability that a bag weighs less than w grams is 0.0849303...

Ten bags of rice are selected at random. Find the probability that exactly one of the bags weighs less than w grams.

The normal probabilities are D5's work and are given here. What is yours is the model: name it, then press one button.

```
qla = ...          # P(exactly 5 of the 12
muffins weigh less than 61 g)
qlb = ...          # P(exactly 1 of the 10 bags
weighs less than w g)

muffins = Bin(12, 0.365112)
bags = Bin(10, 0.0849303)

verify_binomial('1a', qla, P(muffins == 5))
verify_binomial('1b', qlb, P(bags == 1))
```

Theory: a run of values, and moving the boundary yourself

Take $Y \sim B(20, 0.3)$. Draw the values it can take, and mark what each phrase means:

the question says	the event	on the calculator	value
at most 5	$Y \leq 5$	<code>binomcdf(20, 0.3, 5)</code>	0.416
fewer than 5	$Y \leq 4$	<code>binomcdf(20, 0.3, 4)</code>	0.238
at least 5	$Y \geq 5$	<code>1 - binomcdf(20, 0.3, 4)</code>	0.762
more than 5	$Y \geq 6$	<code>1 - binomcdf(20, 0.3, 5)</code>	0.584
at least one	$Y \geq 1$	<code>1 - binompdf(20, 0.3, 0)</code>	0.999

Read the rows as pairs. *At least 5* and *fewer than 5* are complements: together they cover $0, 1, \dots, 20$ once. *More than 5* and *at most 5* are complements. That is the whole rule:

To get "at least k ", subtract "at most $k - 1$ " from one. The boundary

moves by one because k itself belongs to *at least* k , and $1 - P(Y \leq k)$ has thrown it away.

"At least one" is always a complement, and it is the one case where the calculator is not needed: $1 - (0.7)^{20}$. Adding $P(Y = 1) + \dots + P(Y = 20)$ gets the same number more slowly and loses a term more often.

Write the inequality on its own line. $P(X \geq 10) = 1 - P(X \leq 9)$ is a line of working, and the markscheme gives it M1 by itself. The number that follows is the A1.

Task 2 — **May 2024 TZ1 Paper 2 Q6(a)-(b), 4 marks**

In Happyland, the weather on any given day is independent of the weather on any other day. On any day in May, the probability of rain is 0.2. May has 31 days. Find the probability that

(a) it rains on exactly 10 days in May; **(b)** it rains on at least 10 days in May.

Part (c) of this question — the first rainy day being the tenth — is not binomial at all: there is no fixed number of trials. It is in D2.

```
q2a = ...          # P(exactly 10 rainy days)
q2b = ...          # P(at least 10 rainy days)

rain = Bin(31, 0.2)

verify_binomial('2a', q2a, P(rain == 10))
verify_binomial('2b', q2b, P(rain >= 10))
```

Task 3 — **May 2025 TZ2 Paper 2** **Q10(d), 3 marks**

At Adam's Apple Orchard the weights of apples are normally distributed, and an apple is *premium* when its weight is between 170 and 185 grams. Part (c) of the question finds that 62.8364...% of apples are premium.

Boxes are filled with randomly chosen apples. Each box contains 40 apples.

(d) Find the probability that a randomly chosen box contains at least 30 premium apples.

Keep your answer: task 8 is the next part of this question, and it uses it.

```
q3 = ...          # P(at least 30 premium  
apples in a box of 40)  
  
box = Bin(40, 0.628364)  
  
verify_binomial('3', q3, P(box >= 30))
```

Part 2. Describing the distribution, and cutting it

Theory: mean, variance, and working backwards from them

A quiz has 40 multiple-choice questions with four options each, and a student guesses every one. The number right is $X \sim B(40, 0.25)$, and

$$\begin{aligned} E(X) &= np \\ &= 10, \quad \text{Var}(X) \\ &= np(1 - p) \\ &= 40 \times 0.25 \times 0.75 \\ &= 7.5 \end{aligned}$$

$E(X)$ is an **expected number**, not a number that will happen: it does not have to be a whole number, and rounding it to one loses the mark.

A variable built from X . Say each right answer scores +3 and each wrong one -1. The score is $S = 3X - (40 - X) = 4X - 40$, and

$$\begin{aligned}E(S) &= 4E(X) - 40 = 0, \\ \text{Var}(S) &= 4^2 \\ \text{Var}(X) &= 120\end{aligned}$$

The constant goes, the multiplier is squared. Adding 40 to every score moves the distribution without spreading it. Multiplying by 4 spreads it four times — and variance is measured in squares.

Backwards. If $X \sim B(50, p)$ and $\text{Var}(X) = 8$, then $50p(1 - p) = 8$, that is $50p^2 - 50p + 8 = 0$, and

$$p = 0.2 \quad \text{or} \quad p = 0.8$$

Both. $p(1 - p)$ does not change when p and $1 - p$ swap places — counting failures instead of successes spreads out exactly as much. A question that gives only the variance cannot tell the two apart, and neither should your answer.

Task 4 — **May 2021 TZ2 Paper 2 Q10(d)-(e), 6 marks**

The flight times between two cities are normally distributed, and the earlier

parts of the question find that the probability that a randomly selected flight takes more than 80 minutes is 0.0711930...

On a particular day, there are 64 flights scheduled between these two cities.

(d) Find the expected number of flights that will have a flight time of more than 80 minutes. **(e)** Find the probability that more than 6 of the flights on this particular day will have a flight time of more than 80 minutes.

Two "more than"s in one question, and they are not the same kind. The first is inside the definition of a success; the second is the boundary of the event.

```
q4d = ...          # the expected number of
late flights
q4e = ...          # P(more than 6 late
flights)

flights = Bin(64, 0.0711930)

verify_moment('4d', q4d, Expect(flights))
verify_binomial('4e', q4e, P(flights > 6))
```

Task 5 — **November 2023 TZ1**
Paper 2 Q6, 5 marks

The random variable X is such that $X \sim B(25, p)$ and $\text{Var}(X) = 5.75$.

(a) Find the possible values of p .

The random variable Y is such that $Y = 1 - 2X$.

(b) Find $\text{Var}(Y)$.

The check for (a) builds the variance of $B(25, p)$ by adding over all 26 values, solves it itself, and compares both roots with yours. The check for (b) does not use your p at all — it works from the question's own condition, so a slip in (a) costs you (a) only.

```
q5p = [...]          # every possible value of p
q5v = ...             # Var(Y)

condition = Eq(Var(Bin(25, p)), 5.75)

verify_parameter('5a', q5p, condition, p)
verify_moment('5b', q5v, Var(1 - 2 * Bin(25,
p)), given=condition, var=p)
```

Theory: a condition inside one distribution

"Given that..." is D2's conditional probability, unchanged:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

What is new is that A and B are both **ranges of the same X** , and the overlap has to be found on the number line, not on a Venn diagram.

For $Y \sim B(12, 0.4)$:

$$\begin{aligned} P(Y \geq 2 | Y \leq 5) &= \frac{P(2 \leq Y \leq 5)}{P(Y \leq 5)} \\ &= \frac{0.646}{0.665} \\ &= 0.971 \end{aligned}$$

The numerator is not $P(Y \geq 2)$. Values 6, 7, ..., 12 are in $Y \geq 2$ but not in $Y \leq 5$, and they have no business in the fraction. When one event sits entirely inside the other — $P(Y = 3 | Y \leq 4)$ — the overlap is just the smaller event:

$$\begin{aligned}
 P(Y = 3 \mid Y \leq 4) &= \frac{P(Y = 3)}{P(Y \leq 4)} \\
 &= \frac{0.142}{0.438} \\
 &= 0.324
 \end{aligned}$$

Name the events in context. The markscheme writes, every time, *"recognition must be shown in context either in words or symbols but not just $P(A \mid B)$ ".* $P(Y = 3 \mid Y \leq 4)$ is context. $P(A \mid B)$ is not, and scores M0.

The condition has its own boundary, and it is translated exactly as in Part 1. A numerator that is right and a denominator that includes one value too many is a wrong answer by a few per cent — close enough to look fine, and the markscheme has a note naming that exact mistake.

Task 6 — **November 2021 Paper 2 Q3, 7 marks**

A factory manufactures lamps. It is known that the probability that a lamp is

found to be defective is 0.05. A random sample of 30 lamps is tested.

(a) Find the probability that there is at least one defective lamp in the sample.

(b) Given that there is at least one defective lamp in the sample, find the probability that there are at most two defective lamps.

If your (b) comes out bigger than one, the numerator has more in it than the overlap. Look at which values are in both events.

```
q6a = ...          # P(at least one defective)
q6b = ...          # P(at most two defective |
at least one defective)

lamps = Bin(30, 0.05)

verify_binomial('6a', q6a, P(lamps >= 1))
verify_binomial('6b', q6b, P(lamps <= 2,
given=lamps >= 1))
```

Task 7 — November 2023 TZ1 Paper 2 Q10(c), 6 marks

A farmer is growing a field of wheat plants. It is known that the probability

that the height of a plant is greater than 98.1 cm is 0.434.

The farmer measures 100 randomly selected plants. Any plant with a height greater than 98.1 cm is considered ready to harvest. Heights of plants are independent of each other.

(i) Find the probability that exactly 34 plants are ready to harvest. **(ii)** Given that fewer than 49 plants are ready to harvest, find the probability that exactly 34 plants are ready to harvest.

Write down the probability of the condition on its own as well. The markscheme gives it an A1 "seen anywhere" — and it is the number that goes wrong.

```
q7i = ...          # P(exactly 34 ready)
q7l = ...          # P(fewer than 49 ready)
q7ii = ...         # P(exactly 34 ready | fewer
than 49 ready)

plants = Bin(100, 0.434)

verify_binomial('7(i)', q7i, P(plants == 34))
verify_binomial('7(ii) condition', q7l,
P(plants < 49))
verify_binomial('7(ii)', q7ii, P(plants ==
34, given=plants < 49))
```

Part 3. Binomial on top of something, and n as the unknown

Theory: when a success is itself a probability

A seed germinates with probability 0.9. Seeds are sold in packets of 20, and a packet is *good* if at least 18 of its seeds germinate. A shop has 6 packets. What is the probability that exactly 5 are good?

Two binomials, one inside the other, and the whole question is deciding **what one trial is**:

- **inside**, a trial is a seed: $G \sim B(20, 0.9)$, and a packet is good with probability $q = P(G \geq 18) = 0.677$;
- **outside**, a trial is a packet: $K \sim B(6, q)$, and $P(K = 5) = 0.276$.

The mistake is to reach the outside model with the inside $p = B(6, 0.9) =$

which answers a question about six seeds.

Carry q unrounded. It is an answer, so it is tempting to round it; it is also the p of the next model, where the rounding is raised to the fifth power.

"Exactly one of the two." Two people, each with their own model, each succeeding or not independently. It is a tree with two branches, and only the two mixed paths count:

$$\begin{aligned} P(\text{exactly one}) \\ = P(A)(1 - P(B)) + P(B)(1 - P(A)) \end{aligned}$$

At least one would add the path where both succeed. $P(A) + P(B)$ adds it twice.

Task 8 ● — May 2025 TZ2 Paper 2
Q10(e), 2 marks

(This continues task 3: a box of 40 apples, each premium with probability 0.628364...)

(e) If 10 of these boxes are randomly selected, find the probability that exactly

4 boxes have at least 30 premium apples.

Build the outside model from the inside one. The check accepts an answer built from your rounded task 3 — with a remark.

```
q8 = ...          # P(exactly 4 of the 10
boxes have at least 30 premium apples)

box = Bin(40, 0.628364)
boxes = Bin(10, P(box >= 30))      # a
trial is now a box

verify_binomial('8', q8, P(boxes == 4))
```

Task 9 — May 2022 TZ2 Paper 2 Q8, 7 marks

Rachel and Sophia are competing in a javelin-throwing competition. The distances thrown by each are normally distributed, and a single throw reaches 60 m with probability 0.1216725... for Rachel and 0.0824333... for Sophia.

In the first round of competition, each competitor must have five throws. To qualify for the next round of competition, a competitor must record at least one

throw of 60 metres or greater in the first round.

Find the probability that only one of Rachel or Sophia qualifies for the next round of competition.

Seven marks and no parts, so make your own: first the probability that each of them qualifies, then the combination.

```
q9r = ...          # P(Rachel qualifies)
q9s = ...          # P(Sophia qualifies)
q9 = ...           # P(only one of them
qualifies)

rachel = Bin(5, 0.1216725, 'R')      # the
number of Rachel's throws of 60 m or more
sophia = Bin(5, 0.0824333, 'S')

verify_binomial('9 Rachel', q9r, P(rachel >=
1))
verify_binomial('9 Sophia', q9s, P(sophia >=
1))
verify_binomial('9', q9, P((rachel >= 1) ^
(sophia >= 1)))  # ^ is "exactly one of"
```

Theory: the least n , and n from a table

A game gives a prize with probability 0.1 on each go. How many goes make the

probability of at least one prize greater than 0.95?

$$\begin{aligned}P(X \geq 1) &= 1 - 0.9^n > 0.95 \\&\implies 0.9^n < 0.05 \\&\implies n > \frac{\ln 0.05}{\ln 0.9} \\&= 28.43 \dots\end{aligned}$$

and so $n = 29$. Two things to watch in that line:

- **dividing by $\ln 0.9$ flips the inequality**, because $\ln 0.9 < 0$. Do it without flipping and you get $n < 28.43$, which is nonsense for a *least* n ;
- **28.43 is not the answer.** n counts goes. The answer is the whole number on the correct side of the boundary — and checking both neighbours takes ten seconds: $1 - 0.9^{28} = 0.9477$ (not yet), $1 - 0.9^{29} = 0.9529$ (enough).

On a calculator paper you do not need logs. The May 2024 markscheme prints *METHOD 2 (TABLE ONLY APPROACH)* next to the logarithm method, with the same five marks. A

table of $1 - 0.9^n$ against n is also the only way that works when the event is not "at least one".

n from an approximate probability. If $X \sim B(n, 0.1)$ and $P(X \leq 2) \approx 0.485$, no logarithm helps: $P(X \leq 2)$ is three terms, each with its own power of n . Tabulate it — $n = 26$ gives 0.511, $n = 27$ gives 0.485, $n = 28$ gives 0.459 — and read off $n = 27$.

Task 10 — **May 2024 TZ2 Paper 2 Q5, 5 marks**

Consider a random variable X such that $X \sim B(n, 0.25)$.

Determine the least value of n such that $P(X \geq 1) > 0.99$.

The check finds the least n the way the table does: by trying $n = 1, 2, 3, \dots$. If you hand it the boundary from a logarithm, it will tell you why that is not yet the answer.

```
q10 = ...          # the least n

verify_trials('10', q10, lambda n: P(Bin(n,
0.25) >= 1), holds=lambda value: value >
0.99)
```

Task 11 — **May 2025 TZ3 Paper 2 Q11(a)-(b), 8 marks**

Amanda enters data from surveys into a database. It can be assumed that the accuracy of any survey entered is independent of all other surveys entered. From previous records, it is known that Amanda enters 8% of the surveys inaccurately.

(a) On a particular day Amanda enters data from 50 surveys.

(i) Find the probability that Amanda entered at most six surveys inaccurately.

(ii) Given that at most six surveys were entered inaccurately, find the probability that exactly four surveys were entered inaccurately.

On a different day Amanda enters data from n surveys. On this day, the

probability that at most six surveys were entered inaccurately is approximately 0.367.

(b) Find the value of n .

Three techniques in one question: a run of values, a condition, and n from a table. In (b) the answer is not the least n of anything — it is the n whose probability rounds to 0.367.

```
q11a = ...      # P(at most six inaccurate
out of 50)
q11c = ...      # P(exactly four | at most
six)
q11n = ...      # n

surveys = Bin(50, 0.08)

verify_binomial('11a(i)', q11a, P(surveys <=
6))
verify_binomial('11a(ii)', q11c, P(surveys ==
4, given=surveys <= 6))
verify_trials('11b', q11n, lambda n: P(Bin(n,
0.08) <= 6), near=0.367)
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
exact	<i>exactly k</i> : one value, <code>binompdf</code>
cumulative	<i>at most, at least, more than, fewer than</i>
mean	<i>expected number</i> , a mean or a variance, or p from one
conditional	<i>given that</i> , and both events are about the same count
nested	a success is itself a probability, or <i>exactly one of two</i>
trials	n is the unknown

1. A test has 15 questions with four options each, all guessed. Find the probability of exactly 7 right.
2. 5% of bolts are faulty. In a box of 200, find the probability that at most 8 are faulty.
3. $X \sim B(n, p)$ with $E(X) = 12$ and $\text{Var}(X) = 3$. Find n and p .

4. A spinner shows red with probability 0.35 and is spun 20 times. Given that red appears at least 5 times, find the probability that it appears exactly 7 times.
5. Each egg in a carton of 12 is cracked with probability 0.02, and a carton with more than one cracked egg is rejected. Find the probability that exactly 3 of 50 cartons are rejected.
6. How many times must a fair die be rolled for the probability of at least one six to exceed 0.9?
7. A seed germinates with probability 0.85. Of 30 seeds, find the probability that more than 25 germinate.
8. A fair coin is tossed 400 times. Find the standard deviation of the number of heads.
9. Two archers shoot 6 arrows each and hit with probabilities 0.6 and 0.75. Find the probability that exactly one of them hits at least 5 times.
0. $X \sim B(10, 0.3)$. Find $P(X = 2 \mid X < 4)$.

1. A bulb is defective with probability 0.03. In a sample of 25, find the probability that exactly one is defective.
2. $X \sim B(n, 0.2)$ and $P(X \leq 3) \approx 0.411$. Find n .

```
answers = {  
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',  
    7: '', 8: '', 9: '', 10: '', 11: '', 12:  
    '',  
}  
  
trigger_check(answers, {1: 'fa79d4746c21', 2:  
'2dfda8e4279b', 3: '82523faealb0', 4:  
'6929adae851d', 5: '233562de1a02', 6:  
'959a9132f6c7', 7: '2dfda8e4279b', 8:  
'82523faealb0', 9: '233562de1a02', 10:  
'6929adae851d', 11: 'fa79d4746c21', 12:  
'959a9132f6c7'})
```

On the clock — May 2023 TZ1 Paper 2 Q11(c)-(d), 5 marks

Five marks, seven minutes. No hints this time.

A laboratory trial uses an amount of reagent X whose probability density

function gives $P(X < 0.5) = 0.239358\dots$. Each laboratory trial is independent. A trial is considered a success when $X < 0.5$.

(c) Determine the least number of trials required to be 99% sure of at least one success.

Ten trials were conducted.

(d) Find the probability that exactly three trials were successful.

Attempt log

date	time	result

```
qt_n = ...      # the least number of trials
qt_3 = ...      # P(exactly three successes
in ten)

success = 0.239358

verify_trials('timer (c)', qt_n, lambda n:
P(Bin(n, success) >= 1),
             holds=lambda value: value >=
0.99)
verify_binomial('timer (d)', qt_3, P(Bin(10,
success) == 3))
```

Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (a) Let M be the number of the 12 muffins weighing less than 61 g. Twelve independent muffins, each light with the same probability:

$$M \sim B(12, 0.365112\dots), \quad P(M = 5) = 0.213666\dots \\ = \boxed{0.214}$$

1 (b) Let L be the number of the ten bags lighter than w :

$$L \sim B(10, 0.0849303\dots), \quad P(L = 1) = 0.382076\dots \\ = \boxed{0.382}$$

In both, the M1 is for the model. The markscheme accepts it as "*recognition of binomial eg $B(12, 0.365\dots)$* " — written, not implied by the number.

2 (a) Let X be the number of rainy days.
 $X \sim B(31, 0.2)$:

$$\begin{aligned}
 P(X = 10) &= \binom{31}{10} (0.2)^{10} (0.8)^{21} \\
 &= 0.0418894 \dots \\
 &= \boxed{0.0419}
 \end{aligned}$$

2 (b) *At least 10* is everything except $0, \dots, 9$:

$$\begin{aligned}
 P(X \geq 10) &= 1 - P(X \leq 9) \\
 &= 1 - 0.925400 \dots \\
 &= \boxed{0.0746}
 \end{aligned}$$

$1 - P(X \leq 10) = 0.0327$ is the answer to *more than 10*. And the markscheme note is worth reading: "*If no working shown, award (M1)A0 for 0.075 (2 sf)*" — the third figure is not decoration.

3 Let A be the number of premium apples in a box. $A \sim B(40, 0.628364 \dots)$:

$$P(A \geq 30) = 1 - P(A \leq 29) = \boxed{0.0739}$$

The markscheme gives (A1) for the model with both parameters, separately from the M1 for recognising it — three marks on one button.

4 (d) Let L be the number of late flights.
 $L \sim B(64, 0.0711930\dots)$:

$$\begin{aligned} E(L) &= np \\ &= 64 \times 0.0711930\dots \\ &= 4.55635\dots \\ &= \boxed{4.56} \text{ flights} \end{aligned}$$

Not 5, and not 4: an expected number is a mean.

4 (e) *More than 6* starts at 7:

$$\begin{aligned} P(L > 6) &= P(L \geq 7) \\ &= 1 - P(L \leq 6) \\ &= 1 - 0.830880\dots \\ &= \boxed{0.169} \end{aligned}$$

$1 - P(L \leq 5) = 0.304$ counts six late flights as "more than six".

5 (a) $\text{Var}(X) = np(1 - p)$:

$$\begin{aligned}
25p(1 - p) &= 5.75 \\
\implies 25p^2 - 25p + 5.75 & \\
&= 0 \\
\implies p & \\
&= \frac{25 \pm \sqrt{625 - 575}}{50} \\
&= \frac{1}{2} \pm \frac{\sqrt{2}}{10}
\end{aligned}$$

$$\boxed{p = 0.641 \text{ or } p = 0.359}$$

A1A1, one for each. They add to one, and that is not a coincidence: p and $1 - p$ give the same variance.

5 (b) The constant goes, the multiplier is squared:

$$\begin{aligned}
\text{Var}(Y) &= \text{Var}(1 - 2X) \\
&= (-2)^2 \text{Var}(X) \\
&= 4 \times 5.75 \\
&= \boxed{23}
\end{aligned}$$

Whichever p it was — part (b) does not need to know.

6 (a) Let D be the number of defective lamps. $D \sim B(30, 0.05)$:

$$\begin{aligned}P(D \geq 1) &= 1 - P(D = 0) \\&= 1 - 0.95^{30} \\&= \boxed{0.785}\end{aligned}$$

6 (b) Both events are about D . The values that are *at least one* **and** *at most two* are 1 and 2:

$$\begin{aligned}P(D \leq 2 \mid D \geq 1) &= \frac{P(1 \leq D \leq 2)}{P(D \geq 1)} \\&= \frac{0.338903 \dots + 0.258636 \dots}{0.785361 \dots} \\&= \frac{0.597540 \dots}{0.785361 \dots} \\&= \boxed{0.761}\end{aligned}$$

$\frac{P(D \leq 2)}{P(D \geq 1)} = \frac{0.812178}{0.785361} = 1.03$ is the numerator with $D = 0$ still in it — and a probability above one is the loudest signal this topic gives you.

7 (i) Let H be the number of plants ready. $H \sim B(100, 0.434)$:

$$P(H = 34) = 0.0133198 \dots = \boxed{0.0133}$$

7 (ii) *Fewer than 49* is $H \leq 48$:

$$P(H < 49) = P(H \leq 48) = \boxed{0.848}$$

The overlap of *exactly 34* and *fewer than 49* is just $H = 34$:

$$\begin{aligned} P(H = 34 \mid H < 49) &= \frac{0.0133198 \dots}{0.848218 \dots} \\ &= \boxed{0.0157} \end{aligned}$$

The markscheme names the trap in a note: "*If the candidate finds $P(X \leq 49) = 0.890474 \dots$ and uses that to calculate $P(34 \mid X < 49) = 0.0149581 \dots$ award (A0) (M1)(A1)A0.*" One value too many in the denominator costs two of the four marks.

8 A trial is now a box, succeeding with the probability from task 3. Let Y be the number of the ten boxes with at least 30 premium apples:

$$\begin{aligned} Y \sim B(10, 0.0738617 \dots), \quad P(Y = 4) &= 0.00394413 \dots \\ &= \boxed{0.00394} \end{aligned}$$

With 0.0739 carried instead, the answer is 0.00395 — accepted as follow-through, but it is the kind of difference that

sometimes is not. Using $B(10, 0.628)$, the apple's probability instead of the box's, answers a question about ten apples.

9 For one competitor, qualifying means at least one of five throws reaches 60 m. Let $N_R \sim B(5, 0.1216725\dots)$ and $N_S \sim B(5, 0.0824333\dots)$ count their long throws:

$$\begin{aligned} P(N_R \geq 1) &= 1 - (1 - 0.1216725\dots)^5 \\ &= \boxed{0.477} \quad P(N_S \geq 1) \\ &= 1 - (1 - 0.0824333\dots)^5 \\ &= \boxed{0.350} \end{aligned}$$

Only one qualifies — Rachel and not Sophia, or Sophia and not Rachel:

$$\begin{aligned} &0.477264\dots \times 0.650412\dots \\ &+ 0.349588\dots \times 0.522736\dots \\ &= 0.493160\dots = \boxed{0.493} \end{aligned}$$

Or, as the markscheme's second route: $P(A) + P(B) - 2P(A)P(B)$. The 2 is what separates *exactly one* from *at least one*, whose answer 0.660 is one subtraction short.

10 $P(X \geq 1) = 1 - P(X = 0) = 1 - 0.75^n$:

$$1 - 0.75^n > 0.99 \implies 0.75^n < 0.01$$

$$\implies n > \frac{\ln 0.01}{\ln 0.75}$$

$$= 16.0078 \dots$$

$$\boxed{n = 17}$$

The neighbours, as the table method prints them: $n = 16$ gives $0.989977 \dots < 0.99$, and $n = 17$ gives $0.992483 \dots > 0.99$. A boundary of 16.0078 is so close to 16 that rounding to the nearest is the natural mistake, and this question is built to catch it.

11 (a)(i) Let E be the number of inaccurate surveys. $E \sim B(50, 0.08)$:

$$P(E \leq 6) = 0.898128 \dots = \boxed{0.898}$$

11 (a)(ii) *Exactly four* lies inside *at most six*, so the overlap is $E = 4$:

$$\begin{aligned}
 P(E = 4 \mid E \leq 6) &= \frac{P(E = 4)}{P(E \leq 6)} \\
 &= \frac{0.203654 \dots}{0.898128 \dots} \\
 &= \boxed{0.227}
 \end{aligned}$$

11 (b) Now $E \sim B(n, 0.08)$, and $P(E \leq 6)$ falls as n grows. Tabulate around where it crosses 0.367:

n	93	94	95
$P(E \leq 6)$	0.378	0.367	0.356

$$\boxed{n = 94}$$

The markscheme gives M1 for "*at least one correct value of $P(E \leq 6)$ for a value of n , where $n \neq 50$* " — the table is the method.

Timer

(c) Let $Y \sim B(n, 0.239358 \dots)$ count the successes.

$$\begin{aligned}
P(Y \geq 1) &= 1 - (1 - 0.239358 \dots)^n \\
&\geq 0.99 \\
&\implies n \\
&\geq \frac{\ln 0.01}{\ln 0.760641 \dots} \\
&= 16.8321 \dots
\end{aligned}$$

17 trials

By table: $n = 16$ gives $0.987443 \dots$, $n = 17$ gives $0.990448 \dots$. With 0.239 instead of $0.239358 \dots$ the boundary is 16.8612 and the table values 0.987348 , 0.990371 — the markscheme lists both, and the answer does not move.

(d) $Y \sim B(10, 0.239358 \dots)$:

$$P(Y = 3) = 0.242430 \dots = \boxed{0.242}$$